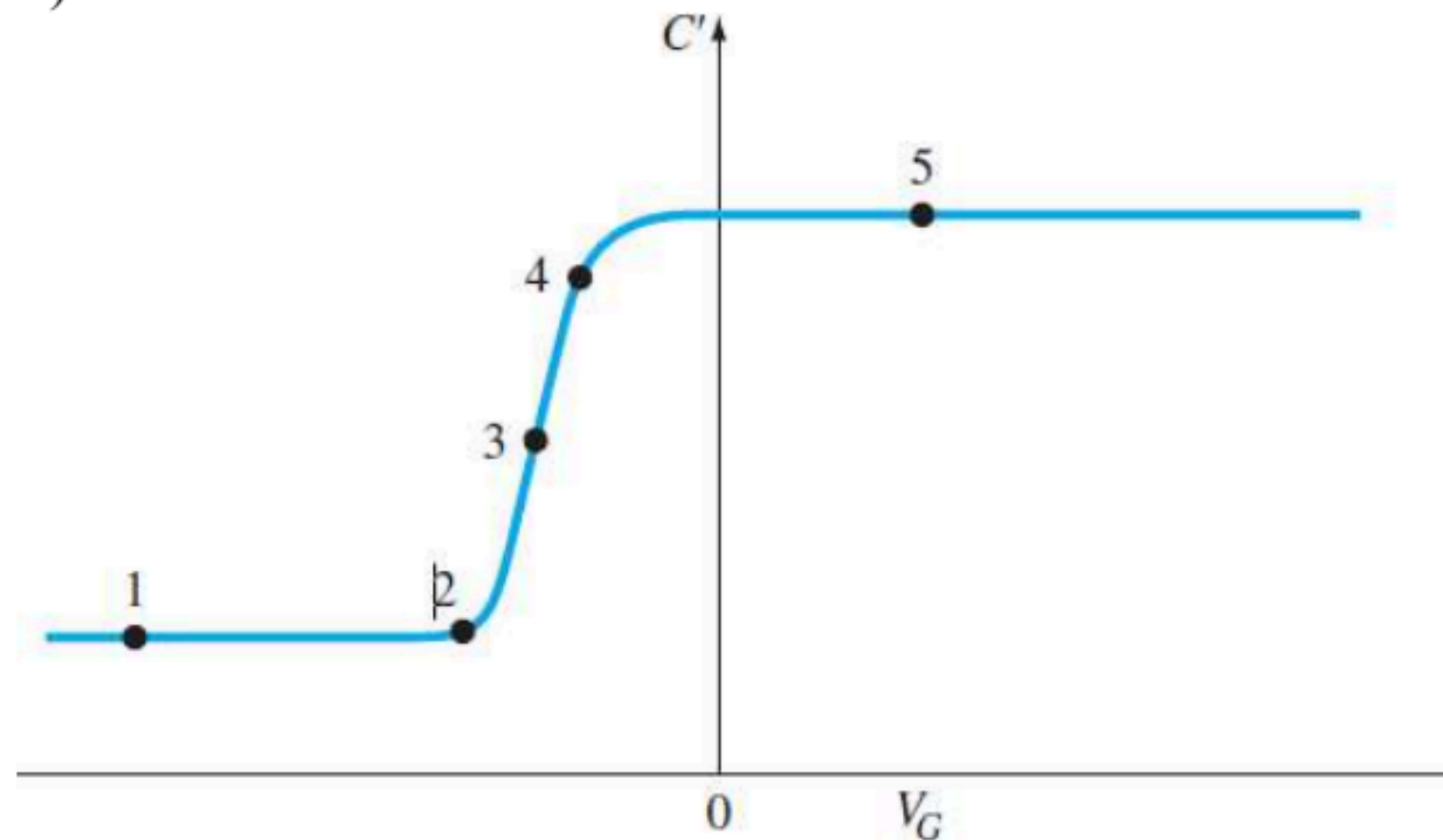


ECE 3103

Homework #9

(10 points)

1. The punch-through voltage of an n-channel MOSFET is to be no less than $V_{DS}=3$ Volts. The source and drain regions are doped $N_d = 10^{19} \text{cm}^{-3}$, and the channel region is doped $N_a = 10^{16} \text{cm}^{-3}$. The source and substrate (body) are at ground potential. Determine the minimum channel (gate) length. (2 points)
2. Read chapter 2 on MOSFETs in “IEEE_Guide_to_State_of_the_Art_Electron_Devices_MOSFETs.pdf”. (On Canvas under Files/Supplements/) and answer the following question: When did J.E. Lilienfeld file his first patent on the idea of a field effect transistor? (3 points)
3. A high-frequency CV characteristic of a MOS structure is shown below. Indicate which points 1, 2, ..5 correspond to flat band, inversion, accumulation, threshold, and depletion modes. (5 points)



1. The punch-through voltage of an n-channel MOSFET is to be no less than $V_{DS}=3$ Volts. The source and drain regions are doped $N_d=10^{19}\text{cm}^{-3}$, and the channel region is doped $N_a=10^{16}\text{cm}^{-3}$. The source and substrate (body) are at ground potential. Determine the minimum channel (gate) length. (2 points)

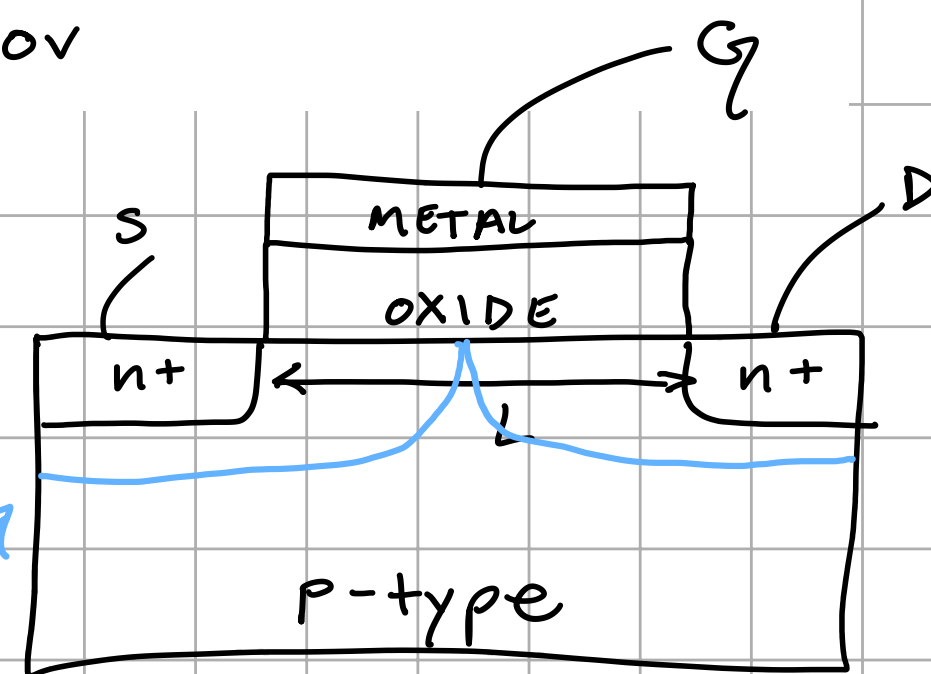
$$V_s = 0\text{V}$$

$$V_{bi} = V_t \ln \left(\frac{N_a N_d}{n_i^2} \right)$$

$$\therefore = (0.0259\text{ V}) \ln \left(\frac{10^{16}\text{cm}^{-3} \cdot 10^{19}\text{cm}^{-3}}{(1.5 \cdot 10^{10}\text{cm}^{-3})^2} \right)$$

assume Si

$$V_{bi} = 0.874\text{ V}$$



Minimum L is longer than source depletion width (x_s) and drain depletion width (x_D).

$$x_D = \sqrt{\frac{2\epsilon(V_{bi} + V_D)}{eN_a}}$$

$$V_{DS} = V_D - 0\text{V} = 3\text{V}$$

$$\therefore = \sqrt{\frac{2(11.7)(8.85 \cdot 10^{-14})(0.874 + 3\text{V})}{(1.6 \cdot 10^{-19})(10^{16}\text{cm}^{-3})}}$$

$$x_D = 7.081 \cdot 10^{-5}\text{ cm}$$

~~$$x_s = x_p = \frac{N_d x_n}{N_a}$$~~

~~$$\therefore = \left(\frac{10^{19}\text{cm}^{-3} (7.081 \cdot 10^{-5})}{10^{16}\text{cm}^{-3}} \right)$$~~

~~$$x_s = 0.07081$$~~

$$x_s = W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d} \right]^{1/2}$$

$$= \sqrt{\frac{2(11.7)(8.85 \cdot 10^{-14})(0.874 + 0\text{V})}{(1.6 \cdot 10^{-19})(10^{19}\text{cm}^{-3})}}$$

$$= 1.06 \cdot 10^{-6}\text{ cm}$$

$$L > x_D + x_s$$

$$L > 7.081 \cdot 10^{-5}\text{ cm} + 1.06 \cdot 10^{-6}\text{ cm}$$

$$L > 7.19 \cdot 10^{-5}\text{ cm}$$

1

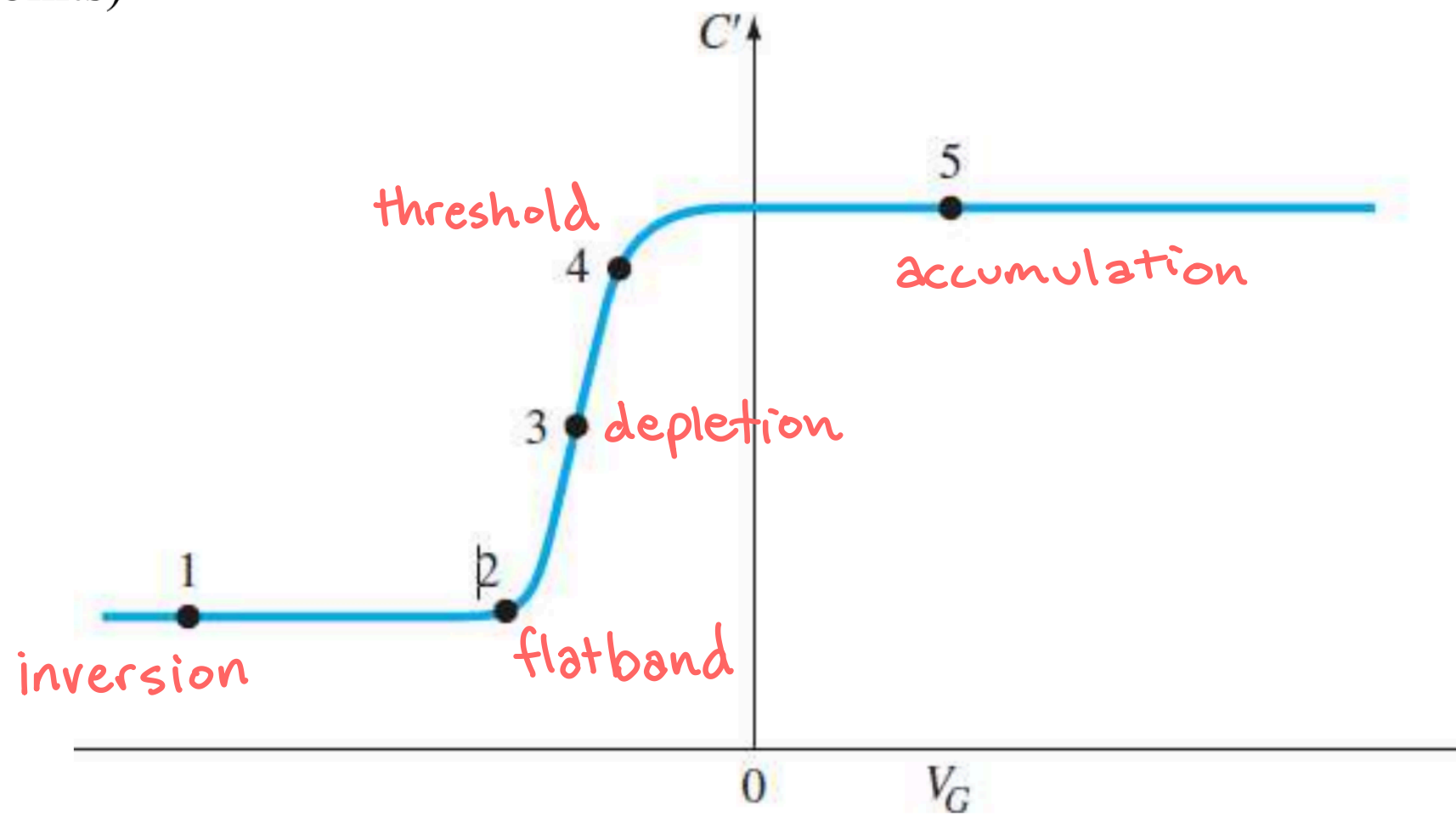
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When did J.E. Lilienfeld file his first patent on the idea of a field effect transistor? (3 points)

[26] J. E. Lilienfeld, “Device for controlling electric current”, U.S. Patent No. 1,900,018, Filed March 28, 1928.

MARCH 28, 1928

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3. A high-frequency CV characteristic of a MOS structure is shown below. Indicate which points 1, 2, ..5 correspond to flat band, inversion, accumulation, threshold, and depletion modes. (5 points)



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